

EduPulse AI

Student Success Early-Warning and Intervention Platform

AI Mini Group Project

Academic prototype localized for a Kabarak University context

Student: Stephen Robert Ochieng Nyambok

Registration number: CS/M/0113/01/24

Lecturer: Dr Andrew Kipkebut

Unit: COMP 413

Submission date: 17 August 2026

Executive Summary / Abstract

EduPulse AI demonstrates an end-of-Semester-1 early-warning workflow using UCI Dataset 697. A leakage-safe seven-input pipeline compared three feature sets and five classifiers, tuned the two strongest candidates and calibrated the selected Support Vector Machine. On the untouched test partition it achieved Macro F1 0.630, balanced accuracy 0.628 and Dropout recall 0.729. The portal translates estimates into supportive review pathways while retaining human control. The source is Portuguese research data, not Kabarak data, so local validation is mandatory.

1. Introduction

Universities need timely, proportionate ways to notice academic or financial difficulty. Machine learning can combine multiple weak signals consistently, but only when the prediction time, data provenance and human-use limits are explicit.

2. Problem Definition

The operational problem is late discovery of students who may benefit from support. The analytical problem is supervised three-class classification of Dropout, Enrolled and Graduate outcomes using only information genuinely available by the end of Semester 1.

3. Project Objectives

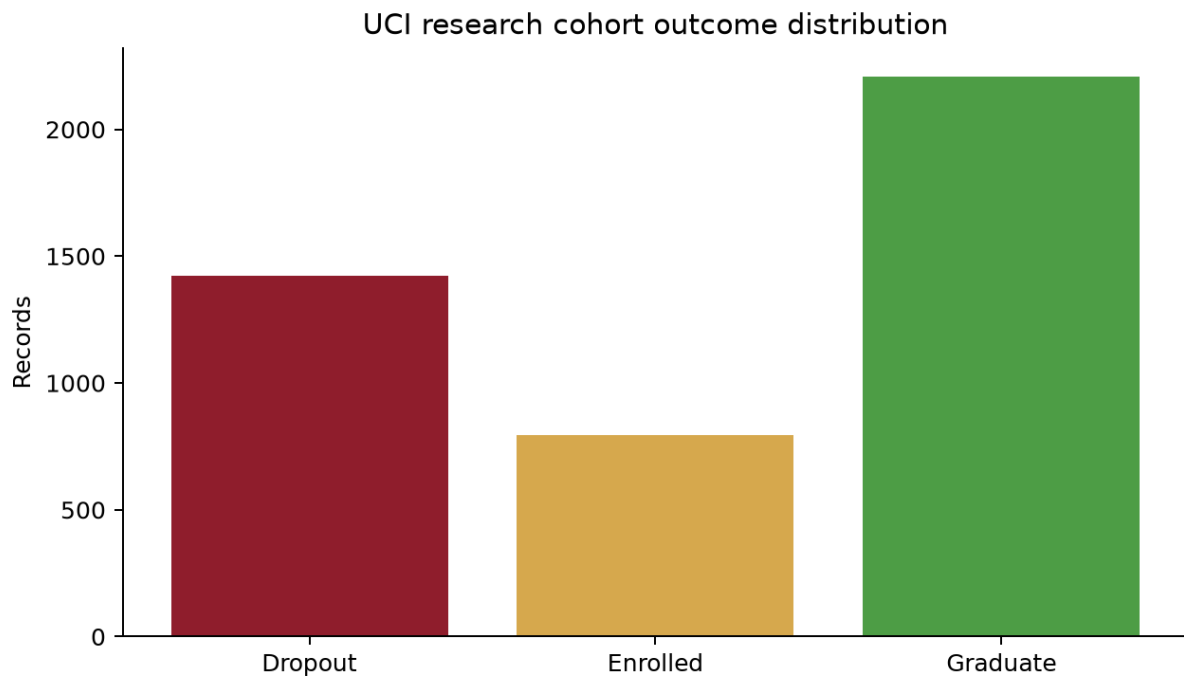
The general objective was to build a reproducible, interpretable early-warning prototype. Specific objectives were to audit public data, prevent Semester 2 leakage, minimize inputs, benchmark models, tune and calibrate probabilities, explain estimates, design supportive workflows and package the result for academic demonstration.

4. Significance / Real-World Relevance

Earlier review can connect a student with an Academic Adviser, Student Finance, Guidance & Counselling or the Dean of Students. EduPulse AI prioritizes attention; it does not automate penalties or replace professional judgement.

5. Dataset & Data Acquisition

The project uses UCI Machine Learning Repository Dataset 697, Predict Students' Dropout and Academic Success, DOI 10.24432/C5MC89, under CC BY 4.0. It has 4,424 Portuguese higher-education records, 36 predictors and three target classes. This is not Kabarak University administrative data.

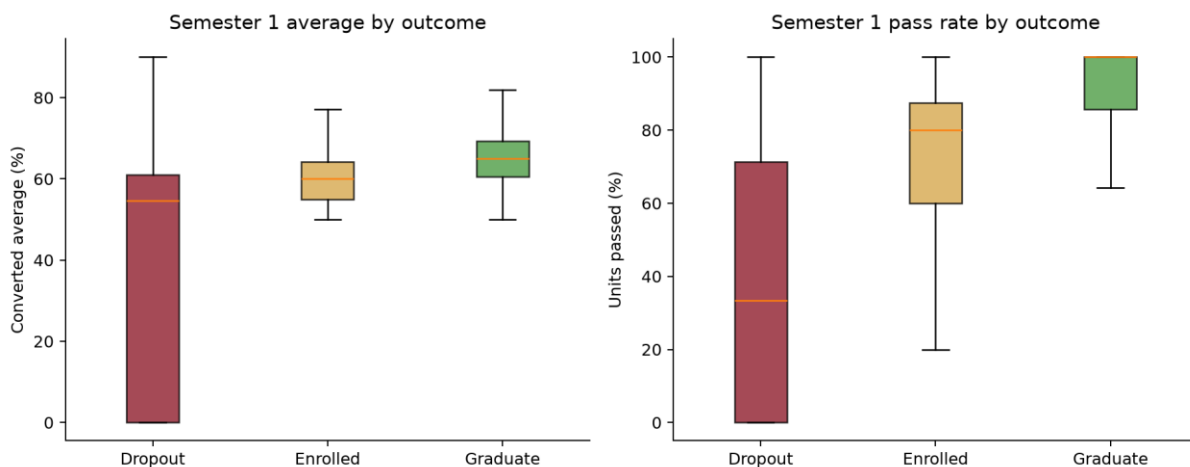


6. Data Cleaning & Preprocessing

Automated checks found 0 missing cells, 0 duplicate rows, valid target labels and numeric source predictors. Range and category checks found zero invalid binary cells, zero negative Semester 1 count cells and zero grades outside 0–20. Headers were normalized. Median imputation remains inside the pipeline as a defensive deployment safeguard, followed by standardization. Raw data are preserved unchanged and identified by SHA-256.

7. Exploratory Data Analysis

The target is imbalanced, with Graduate the largest and Enrolled the smallest class. Semester 1 average and pass rate show clear shifts by final outcome but substantial overlap. Fee status is associated with outcomes. These patterns justify multivariable classification but do not establish causation.



8. Feature Engineering

Seven UI fields are transformed inside the saved sklearn pipeline into ten model features. `pass_rate` equals `units_passed` divided by `units_registered`; `units_not_passed` is their non-negative difference; `assessments_per_unit` is guarded against division by zero. The UCI 0–20 grade is multiplied by five for an exact 0–100 display conversion.

feature_set	input_count	macro_f1	balanced_accuracy	dropout_recall
C — engineered compact inputs	7	0.662	0.669	0.653
B — compact raw inputs	7	0.657	0.663	0.614
A — wider leakage-safe Semester 1	13	0.652	0.656	0.653

9. Model Development

The training partition supported a fair feature-set experiment and five-model benchmark. Logistic regression, random forest, extra trees, support vector machine and histogram gradient boosting were evaluated with five-fold stratified cross-validation. Macro F1 was primary, supplemented by balanced accuracy, accuracy, weighted F1 and Dropout recall.

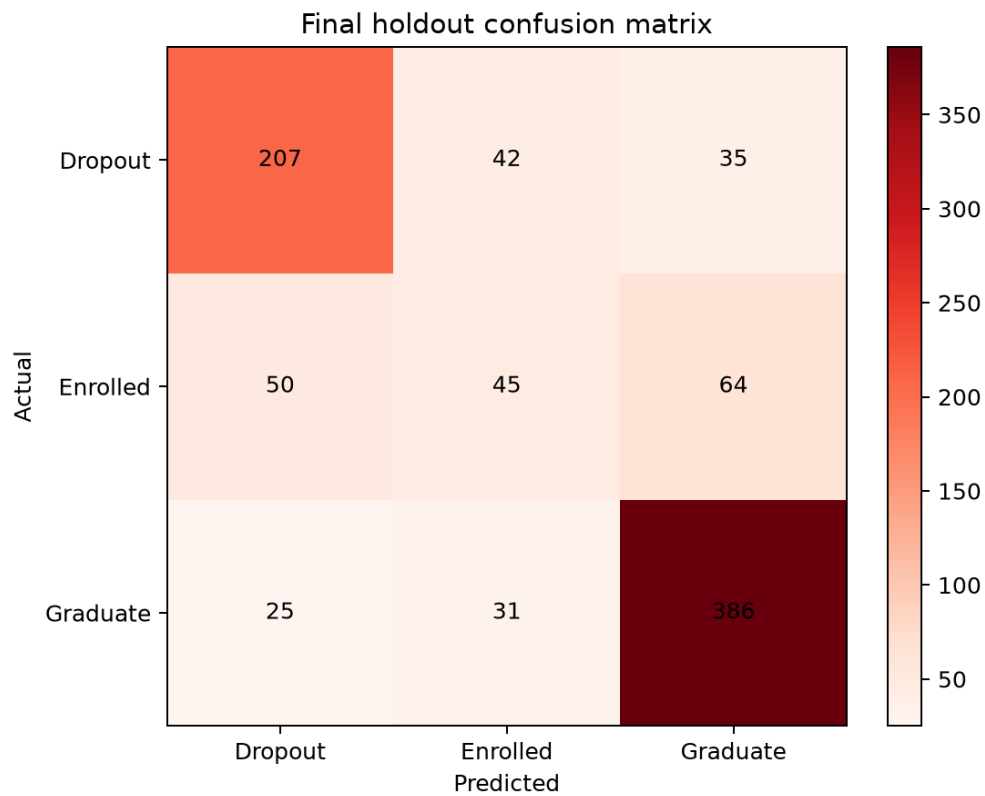
model	macro_f1	balanced_accuracy	accuracy	dropout_recall
Support Vector Machine	0.664	0.677	0.700	0.624
Logistic Regression	0.662	0.669	0.705	0.653
Extra Trees	0.657	0.661	0.705	0.691
Random Forest	0.642	0.642	0.702	0.707
Histogram Gradient Boosting	0.627	0.630	0.680	0.687

10. Hyperparameter Tuning

The two strongest baseline algorithms were tuned with bounded randomized searches on training folds only. Support Vector Machine achieved the strongest tuned CV Macro F1 of 0.665, with parameters `{"model__gamma": "scale", "model__C": 8.0}`.

11. Model Evaluation

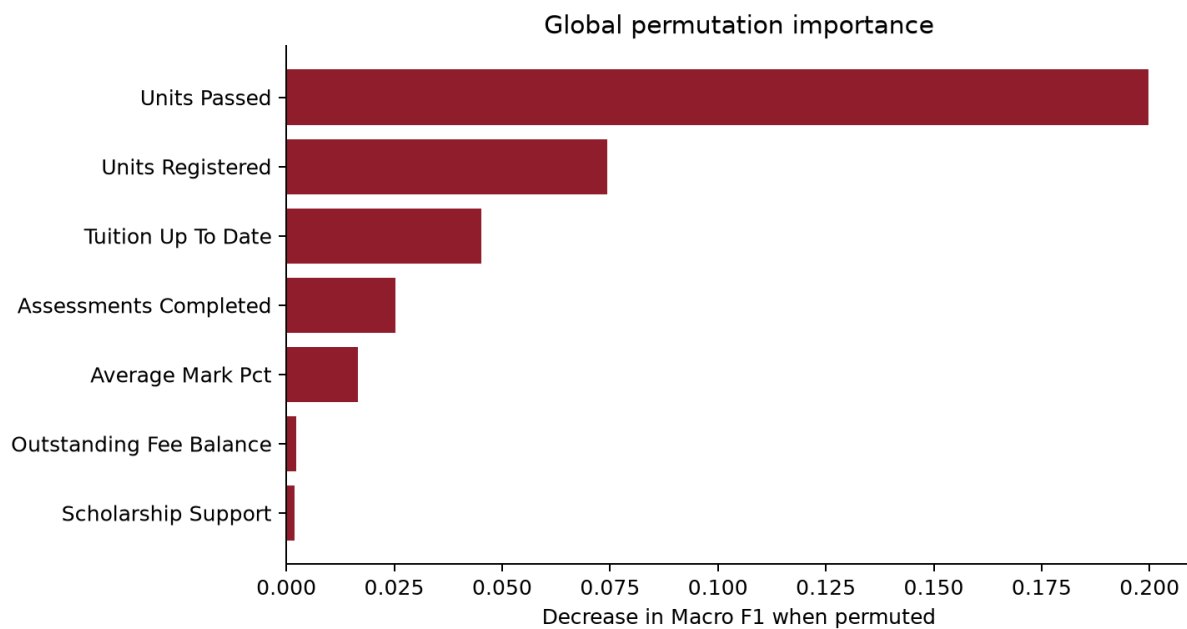
The untouched 885-record test set was evaluated once after selection. Accuracy was 0.721, balanced accuracy 0.628, Macro F1 0.630, weighted F1 0.709, log loss 0.674 and weighted multiclass ROC-AUC 0.867.



class	precision	recall	f1	support
Dropout	0.734	0.729	0.731	284
Enrolled	0.381	0.283	0.325	159
Graduate	0.796	0.873	0.833	442

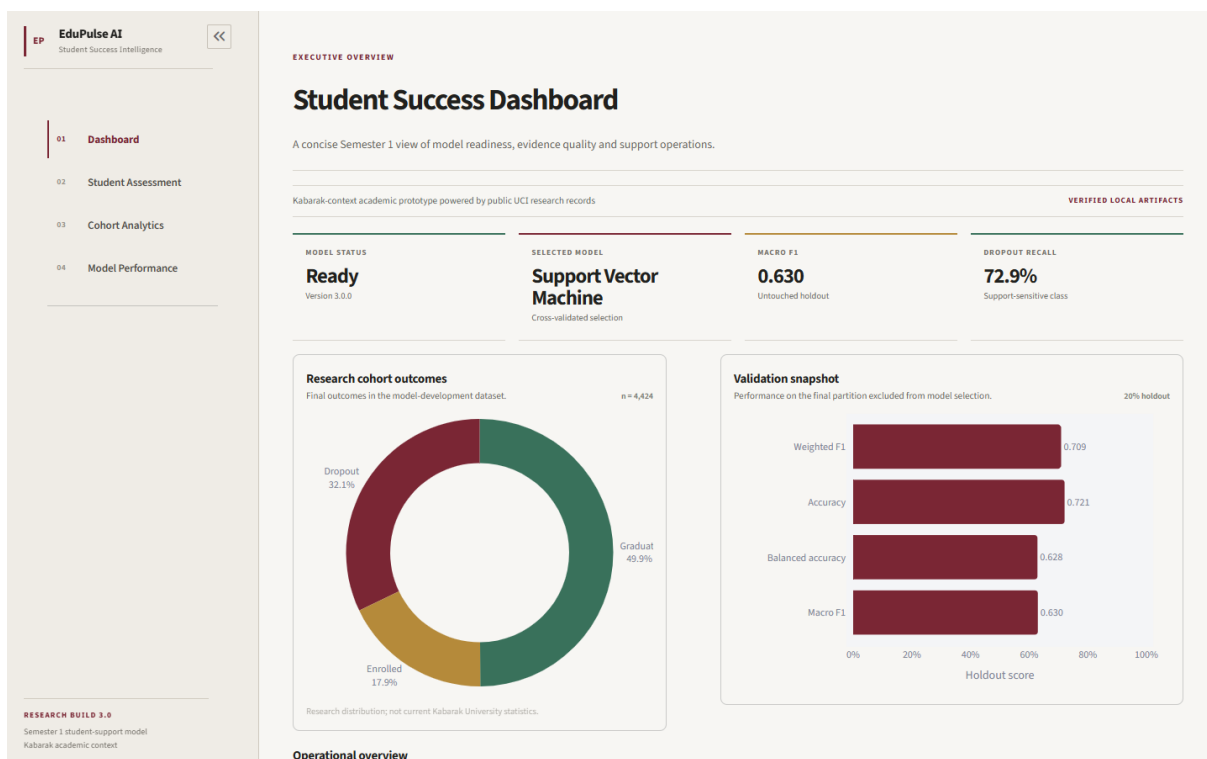
12. Results & Interpretation

Dropout precision was 0.734, recall 0.729 and F1 0.731. Enrolled remained the hardest class. Permutation importance identifies influential inputs but must not be interpreted as causal evidence.



13. EduPulse AI System Design

The Streamlit portal contains Dashboard, Student Assessment, Cohort Analytics and Model Performance pages. The assessment form includes optional reference fields outside the model contract, three synthetic demo profiles, all outcome probabilities, local sensitivity signals, supportive pathways and a downloadable summary. Cohort Analytics includes validated CSV batch assessment.



14. Deployment

A full Windows launcher creates the environment, installs changed requirements, fetches data if absent, retrains only when artifacts are missing or outdated and starts Streamlit. A fast-start launcher uses existing artifacts without retraining. The local demonstration has no internet dependency after setup. Docker configuration is also provided.

15. Responsible AI, Ethics & Limitations

The project minimizes sensitive inputs, forbids identity and Semester 2 data, uses supportive language and requires human review. Important limitations are geographic context mismatch, no Kabarak validation, outcome ambiguity, model error, proxy effects in financial fields and possible distribution shift.

16. Recommendations

Before operational use: establish governance and lawful basis; complete a data-protection impact assessment; map and quality-check local Semester 1 data; validate retrospectively and prospectively; assess calibration and fairness; train staff; communicate with students; log access; monitor drift; and provide correction and appeal mechanisms.

17. Conclusion

EduPulse AI satisfies the academic goal with a real dataset, reproducible ML methodology, strict leakage control, compact interface, transparent evidence and presentation-ready documentation. Its responsible contribution is decision support for early help—not automated judgement.

References

Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2021). Predicting student dropout and academic success. *Data*, 7(11), 146. <https://doi.org/10.3390/data7110146>

UCI Machine Learning Repository. Predict Students' Dropout and Academic Success (Dataset 697). <https://doi.org/10.24432/C5MC89>. CC BY 4.0.

scikit-learn documentation. <https://scikit-learn.org/stable/>